SWITCHING AND VLAN

1. Simple VLAN Network Diagram

```
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       |          Switch             |
        ------------------------------
      /              |                \
     /               |                 \
    /                |                  \
 Laptop          Phone            Smart TV
 VLAN 20         VLAN 20           VLAN 10
(Personal)      (Personal)      (Streaming)
```

- VLAN 10 = Streaming devices (Smart TV)
- VLAN 20 = Personal devices (Laptop, Phone)

2. Create VLANs and Assign PCs

Switch Configuration

```
enable
configure terminal

vlan 10
name STREAMING

vlan 20
name PERSONAL

interface range fa0/1-3
switchport mode access
switchport access vlan 10

interface range fa0/4-5
switchport mode access
switchport access vlan 20

end
```

PC Assignment

PC	Port	VLAN
PC1	Fa0/1	10
PC2	Fa0/2	10
PC3	Fa0/3	10
PC4	Fa0/4	20
PC5	Fa0/5	20

Verification

```
show vlan brief
```

Ping Test

- PC1 → PC2 ✅ Success
- PC1 → PC4 ❌ Fail
- PC2 → PC5 ❌ Fail

Different VLANs cannot communicate without routing.

3. Move a PC from VLAN 10 to VLAN 20

Before

interface fa0/3
switchport mode access
switchport access vlan 10
PC3 belongs to VLAN 10.

After

interface fa0/3
switchport mode access
switchport access vlan 20
PC3 now belongs to VLAN 20.

Connectivity Test

Before

- PC3 → PC1 ✅ Success
- PC3 → PC4 ❌ Fail

After

- PC3 → PC1 ❌ Fail
- PC3 → PC4 ✅ Success

4. Automate Assigning 10 Ports to VLAN 30

enable
configure terminal
This uses the interface range command, so you do not need to configure all 10 ports individually.